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NOETHER'S THEOREM AND THE CASIMIR HONEYCOMB DRIVE: ON NON-CONSERVATION OF MOMENTUM IN ASYMMETRIC VACUUM SYSTEMS

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In this paper the Casimir honeycomb drive is analyzed with the help of Noether's theorem. The system under consideration - a perfectly conducting plate with a nanoscale honeycomb structure on one side - lacks translational symmetry along the axis normal to the plate due to the geometric asymmetry of vacuum modes on its sides. As a consequence, the vacuum Hamiltonian depends explicitly on the coordinate l (the effective cavity length), and by Noether's theorem, the corresponding component of momentum is not conserved. This non-conservation manifests as a net quantum-vacuum force (Casimir thrust). It is shown that the derived expression for the force per unit area directly reflects this broken symmetry and is consistent with both Hamiltonian approach and the difference in the energy density. The result underscores a profound physical principle: macroscopic thrust from quantum vacuum fluctuations arises because the system is not invariant under spatial translations.

Keywords: *Noether's theorem, Casimir effect, quantum vacuum, non-conservation of momentum, nanohoneycomb, Casimir thrust*

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INTRODUCTION

Noether's theorem establishes a profound correspondence between continuous symmetries of a physical system and conserved values of quantities. In particular, invariance under spatial translations implies conservation of linear momentum. In the classical two-plate Casimir configuration, the system remains symmetric under translations perpendicular to the plates, so transverse momentum is conserved; only the normal component is affected by boundary conditions.

In contrast, the *one-body* Casimir honeycomb drive breaks translational symmetry explicitly along the z-axis due to different boundary conditions on the different sides of a single plate: one side being smooth and another structured with conductive nanocavities. This asymmetry renders the vacuum Hamiltonian $\langle 0|\hat{\mathscr{H}}|0\rangle$ a function of the geometric parameter l, which represents the effective length of the nanocavities.

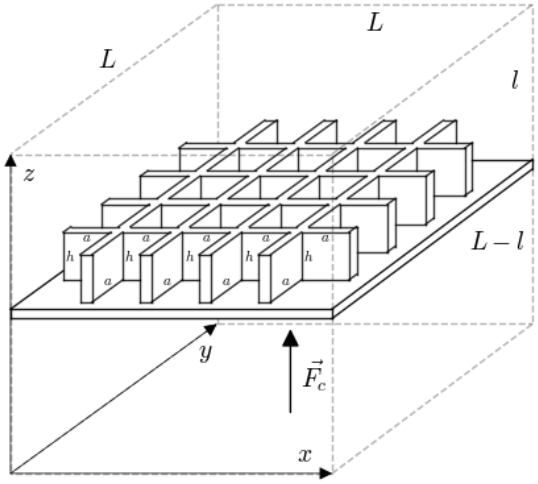


Fig.1. Cubic cavity with a plate covered by honeycomb.

NOETHER'S THEOREM AND MOMENTUM NON-CONSERVATION

Let us consider the vacuum Hamiltonian per unit area derived for the honeycomb system:

$$\frac{\langle 0|\hat{\mathscr{H}}|0\rangle}{S} = \frac{\hbar c}{a^2\pi}\left\{ l \sum_{n_x=(0)1}^{\infty}\sum_{n_y=(0)1}^{\infty}\int_0^\infty \sqrt{\frac{\pi^2}{a^2}(n_x^2+n_y^2)+k_z^2}\,dk_z + (L-l)\int_0^\infty\int_0^\infty\int_0^\infty \sqrt{k_x^2+k_y^2+k_z^2}\,dk_z\,\frac{a}{\pi}dk_x\frac{a}{\pi}dk_y \right\}.$$

This expression depends explicitly on l. Hence, the system is **not invariant** under infinitesimal translations $z \to z+\delta z$.

According to Noether's theorem, the generator of spatial translations—momentum component P_z — is conserved **if and only if** the Hamiltonian is independent of z. Since $\partial\langle H\rangle/\partial l \neq 0$, the symmetry is broken, and P_z is not conserved.

The resulting force is

$$\frac{F}{S} = \frac{\partial}{\partial l}\frac{\langle 0|\hat{\mathscr{H}}|0\rangle}{S} = \frac{\hbar c}{a^2\pi}\left\{ \sum_{n_x=(0)1}^{\infty}\sum_{n_y=(0)1}^{\infty}\int_0^\infty \sqrt{\frac{\pi^2}{a^2}(n_x^2+n_y^2)+k_z^2}\,dk_z - \int_0^\infty\int_0^\infty\int_0^\infty \sqrt{k_x^2+k_y^2+k_z^2}\,dk_z\,dn_x dn_y \right\},$$

which matches the Casimir thrust formula previously obtained via the difference in energy density.

Thus, the appearance of net thrust is not unexpected - it is the consequence of a broken translational symmetry. The quantum vacuum "pushes" more strongly on one side because the quantun vacuum oscillation mode structure differs, and Noether's theorem provides the fundamental explanation why momentum does not need to be conserved.

THE STRESS TENSOR FROM THE CAVITY MODE FIELDS: SURFACE INTEGRALS AND THE JUMP AT THE METAL

In this section the Maxwell stress tensor is computed directly from the fields of the cavity eigenmodes, without assuming the momentum flux of a plane wave. Gaussian units and Tamm's notation are used.

1. Geometry. A plate of zero thickness lies in the plane $z = 0$ inside a cube L^3 with perfectly conducting walls. On the side $z > 0$ the plate carries perfectly conducting cells of cross-section $a \times a$ whose walls reach the cube face $z = l$ (in [1] $l = L/2$). On the side $z < 0$ there is the smooth face of the plate and the large cavity $L \times L \times (L-l)$. Each cavity is a rectangular resonator $a \times b \times h$: for a cell $a = b$, $h = l$; for the large cavity $a = b = L$, $h = L - l$. In the formulas below z is measured from the resonator face that coincides with the plate (for the large cavity this is $-z$).

2. Mode fields. Let $k_x = \pi n_x/a$, $k_y = \pi n_y/b$, $k_z = \pi n_z/h$, $K^2 = k_x^2 + k_y^2$, $k^2 = K^2 + k_z^2$, $\omega = ck$. TM mode ($H_z = 0$, $n_x, n_y \geq 1$, $n_z \geq 0$):

$$E_z = A\sin k_x x \sin k_y y \cos k_z z \cos\omega t, \qquad H_z = 0,$$
$$E_x = -A\frac{k_x k_z}{K^2}\cos k_x x \sin k_y y \sin k_z z \cos\omega t, \quad H_x = -A\frac{k\,k_y}{K^2}\sin k_x x \cos k_y y \cos k_z z \sin\omega t,$$
$$E_y = -A\frac{k_y k_z}{K^2}\sin k_x x \cos k_y y \sin k_z z \cos\omega t, \quad H_y = A\frac{k\,k_x}{K^2}\cos k_x x \sin k_y y \cos k_z z \sin\omega t.$$

TE mode ($E_z = 0$, $n_z \geq 1$, $(n_x, n_y) \neq (0,0)$):

$$H_z = A\cos k_x x \cos k_y y \sin k_z z \cos\omega t, \qquad E_z = 0,$$
$$H_x = -A\frac{k_x k_z}{K^2}\sin k_x x \cos k_y y \cos k_z z \cos\omega t, \quad E_x = -A\frac{k\,k_y}{K^2}\cos k_x x \sin k_y y \sin k_z z \sin\omega t,$$
$$H_y = -A\frac{k_y k_z}{K^2}\cos k_x x \sin k_y y \cos k_z z \cos\omega t, \quad E_y = A\frac{k\,k_x}{K^2}\sin k_x x \cos k_y y \sin k_z z \sin\omega t.$$

3. Check of the fields. By substitution, $\nabla\cdot\mathbf{E} = \nabla\cdot\mathbf{H} = 0$, $\nabla\times\mathbf{E} = -\frac{1}{c}\partial_t\mathbf{H}$, $\nabla\times\mathbf{H} = \frac{1}{c}\partial_t\mathbf{E}$ for $\omega = ck$, and the perfect-conductor boundary conditions hold: tangential $\mathbf{E}$ and normal $\mathbf{H}$ vanish on all six faces (at $x = 0, a$: $E_y = E_z = 0$, $H_x = 0$; at $y = 0, b$: $E_x = E_z = 0$, $H_y = 0$; at $z = 0, h$: $E_x = E_y = 0$, $H_z = 0$). These relations were also checked numerically for the modes $(2,1,3)$ with $a : b : h = 1 : 1.3 : 2.7$ (residual of order 10^{-9}).

4. The tensor component. Following Tamm, $T_{ik} = \frac{1}{4\pi}(E_iE_k + H_iH_k) - \frac{\delta_{ik}}{8\pi}(E^2 + H^2)$, and the force on the bodies inside a closed surface is $\oint T_{ik} n_k\, dS$ (outward normal). Hence

$$T_{zz} = \frac{1}{8\pi}\left(E_z^2 + H_z^2 - E_x^2 - E_y^2 - H_x^2 - H_y^2\right).$$

The momentum flux is $\Pi_{ik} = -T_{ik}$; the pressure on a horizontal element is denoted $p \equiv -\langle T_{zz}\rangle$.

5. Averaging over the cross-section and the period. We average over the area $S = ab$ and over the period $2\pi/\omega$. For $n_x, n_y \geq 1$ each factor $\sin^2$ or $\cos^2$ in x, in y and in t gives $\frac{1}{2}$. For the TM mode:

$$\langle E_z^2\rangle = \frac{A^2}{8}\cos^2 k_z z, \qquad \langle E_x^2 + E_y^2\rangle = \frac{A^2}{8}\frac{k_x^2 + k_y^2}{K^4}k_z^2\sin^2 k_z z = \frac{A^2}{8}\frac{k_z^2}{K^2}\sin^2 k_z z, \qquad \langle H_x^2 + H_y^2\rangle = \frac{A^2}{8}\frac{k^2}{K^2}\cos^2 k_z z.$$

$$8\pi\langle T_{zz}\rangle = \frac{A^2}{8}\left[\cos^2 k_z z\left(1 - \frac{k^2}{K^2}\right) - \frac{k_z^2}{K^2}\sin^2 k_z z\right] = -\frac{A^2}{8}\frac{k_z^2}{K^2}\left(\cos^2 k_z z + \sin^2 k_z z\right) = -\frac{A^2}{8}\frac{k_z^2}{K^2},$$

where $1 - k^2/K^2 = -k_z^2/K^2$ was used. The energy density $w = (E^2 + H^2)/8\pi$ of the same mode is $8\pi\langle w\rangle(z) = \frac{A^2}{8}\left[\cos^2 k_z z + \frac{k_z^2}{K^2}\sin^2 k_z z + \frac{k^2}{K^2}\cos^2 k_z z\right]$; averaging also over z (for $n_z \geq 1$) gives $8\pi\overline{w} = \frac{A^2}{16}\cdot\frac{K^2 + k_z^2 + k^2}{K^2} = \frac{A^2}{8}\frac{k^2}{K^2}$. Therefore

$$p = -\langle T_{zz}\rangle = \overline{w}\,\frac{k_z^2}{k^2}.$$

For the TE mode the same steps give $\langle H_z^2\rangle = \frac{A^2}{8}\sin^2 k_z z$, $\langle H_x^2 + H_y^2\rangle = \frac{A^2}{8}\frac{k_z^2}{K^2}\cos^2 k_z z$, $\langle E_x^2 + E_y^2\rangle = \frac{A^2}{8}\frac{k^2}{K^2}\sin^2 k_z z$, hence $8\pi\langle T_{zz}\rangle = -\frac{A^2}{8}\frac{k_z^2}{K^2}$ and the same result $p = \overline{w}\,k_z^2/k^2$. If one transverse index is zero (possible only for TE), the corresponding $\cos^2$ equals 1 instead of $\frac{1}{2}$; the common factor doubles both in the pressure and in the energy, and the ratio is unchanged. The TM mode with $n_z = 0$ has $k_z = 0$ and exerts no pressure.

6. Independence of z. In item 5 the z-dependence dropped out ($\cos^2 + \sin^2 = 1$). The general reason: in vacuum $\partial T_{zk}/\partial x_k = \frac{1}{4\pi c}\partial_t[\mathbf{E}\times\mathbf{H}]_z$, and the period average of a time derivative vanishes. Integrating over the cross-section leaves the fluxes through the side walls: $T_{zx} = \frac{1}{4\pi}(E_zE_x + H_zH_x)$, and on the walls $x = 0, a$ we have $E_z = 0$ (tangential field) and $H_x = 0$ (normal field), so $T_{zx} = 0$; likewise for T_{zy} at $y = 0, b$. Hence $\frac{d}{dz}\int\langle T_{zz}\rangle\, dx\, dy = 0$. Note that only the cross-section integral is independent of z; at a given point (x,y), T_{zz} does vary with z. The force involves the integral.

7. Zero-point oscillations. Each mode is an oscillator; in the ground state its energy is $\hbar\omega/2$, with equal electric and magnetic parts $\hbar\omega/4$. This is exactly the distribution given by the classical period average of item 5 if the amplitude is chosen so that $\overline{w}V = \hbar\omega/2$. The contribution of a mode to the pressure is therefore $p_{\text{mode}} = \frac{\hbar\omega}{2V}\frac{k_z^2}{k^2}$, and the pressure in a cavity is

$$p = \frac{\hbar c}{2V}\sum_{\text{modes}}\frac{k_z^2}{k},$$

the sum running over TM and TE modes with the index ranges of item 2. It differs from the energy sum, eq. (1) of [1], only by the summand: k_z^2/k instead of k.

Identity with the Hamiltonian approach. For $k_z = \pi n_z/h$,

$\partial k/\partial h = -k_z^2/(kh)$, and for each mode

$$-\frac{\partial}{\partial h}\frac{\hbar c k}{2} = \frac{\hbar c}{2h}\frac{k_z^2}{k} = S\,p_{\rm mode}, \qquad V = Sh.$$

The force obtained from the stress tensor coincides with $-\partial E/\partial h$ mode by mode, before any regularization.

8. Surfaces $z = \text{const}$ for the honeycomb plate. From items 5–7:

- $z \to 0^+$ (vacuum at the cell bottom), $z = d$ with $0 < d < l$, $z \to l^-$ (at the auxiliary cube wall): $-\langle T_{zz}\rangle = p_{\rm cell} = \frac{\hbar c}{2a^2 l}\sum_{\rm cell}\frac{k_z^2}{k}$;
- $z \to 0^-$ (vacuum at the smooth face), $z = -d'$ with $0 < d' < L - l$, $z \to -(L-l)^+$: $-\langle T_{zz}\rangle = p_{\rm big} = \frac{\hbar c}{2L^2(L-l)}\sum_{\rm large}\frac{k_z^2}{k}$;
- inside the metal of the plate and of the walls: $T_{ik} = 0$.

The plane $z = d$ in the cell region crosses the cell walls; for walls of zero thickness they do not contribute to the integral, and the cells cover the whole plate area.

9. The jump at the metal surface. At the face $z = 0$ (vacuum outside, metal inside with zero fields) the boundary conditions require continuity of tangential $\mathbf{E}$ and normal $\mathbf{H}$; on the vacuum side they vanish: $E_x = E_y = 0$, $H_z = 0$. Normal $\mathbf{E}$ and tangential $\mathbf{H}$ jump by $4\pi\sigma$ and $\frac{4\pi}{c}\mathbf{i}$ respectively and are non-zero on the vacuum side. Hence

$$T_{zz}\big|_{\rm vacuum} = \frac{1}{8\pi}\left(E_z^2 - H_x^2 - H_y^2\right), \qquad T_{zz}\big|_{\rm metal} = 0.$$

Explicitly, for the TM mode at $z = 0$: $\langle E_z^2\rangle = \frac{A^2}{8}$, $\langle H_x^2 + H_y^2\rangle = \frac{A^2}{8}\frac{k^2}{K^2}$, so $-\langle T_{zz}\rangle(0^+) = \frac{A^2}{64\pi}\left(\frac{k^2}{K^2} - 1\right) = \frac{A^2}{64\pi}\frac{k_z^2}{K^2} = \overline{w}\frac{k_z^2}{k^2}$, the same value as in the bulk of the cavity (item 6). For the TE mode at $z = 0$, $E_z = 0$ and $-\langle T_{zz}\rangle(0^+) = \frac{1}{8\pi}\langle H_x^2 + H_y^2\rangle = \overline{w}k_z^2/k^2$. The jump $[\![T_{zz}]\!] = T_{zz}|_{\rm metal} - T_{zz}|_{\rm vacuum} = p$ is the surface force per unit area acting on the surface charges $\sigma = E_n/4\pi$ in the field E_n and on the surface currents $\mathbf{i} = \frac{c}{4\pi}[\mathbf{n}\times\mathbf{H}]$ in the field $\mathbf{H}_t$:

$$\frac{f_z}{S} = \frac{1}{8\pi}\left(\langle H_t^2\rangle - \langle E_n^2\rangle\right).$$

In the vacuum state $\langle\sigma\rangle = 0$ and $\langle\mathbf{i}\rangle = 0$, but the correlations $\langle\sigma E_n\rangle$ and $\langle\mathbf{i}\times\mathbf{H}\rangle$ do not vanish: the force is quadratic in the fields. For TE modes there is no surface charge at all on the xy faces ($E_n = E_z = 0$), and the tensor jump is produced by the surface currents alone. This is an explicit example of the case pointed out in the remark on Tamm's § 33: a jump of the stress tensor at a surface that carries no charge. For TM modes the fluctuating charges reduce the pressure by $\langle E_n^2\rangle/8\pi$.

10. The force on the plate. Take a closed surface made of the faces $z = z_1$ ($0 < z_1 \le l$) and $z = -z_2$ ($0 < z_2 \le L - l$) spanning the whole cross-section $L \times L$; its lateral faces lie on the cube walls, where $T_{zx} = T_{zy} = 0$ (item 6). Then

$$F_z = \left[\langle T_{zz}\rangle(z_1) - \langle T_{zz}\rangle(-z_2)\right]L^2 = (p_{\rm big} - p_{\rm cell})\,L^2,$$

independent of z_1, z_2. For $z_1, z_2 \to 0$ this is the sum of the two jumps at the faces of the plate (the force is applied to the metal surfaces); for $z_1 = l$, $z_2 = L - l$ these are the faces at the auxiliary cube walls (for $l = L/2$, the faces $z = \pm L/2$ above); intermediate positions give the same. $F_z > 0$ (a push towards the cells) when $p_{\rm big} > p_{\rm cell}$.

11. Check on Casimir's two plates. Plates at $z = 0$ and $z = a$, outer cavities up to $z = \pm L/2$. In the gap $k_z = \pi n/a$ and the transverse spectrum is continuous; per unit area (two polarizations)

$$p_{\rm in} = \frac{\hbar c}{a}\sum_{n\ge 1}\int\frac{d^2k_\parallel}{(2\pi)^2}\frac{k_z^2}{k}$$

(the $n = 0$ mode has $k_z = 0$ and gives no pressure), and outside the sum over n is replaced by $\frac{a}{\pi}\int dk_z$. By item 6, $p_{\rm in}$ is the same at the centre of the gap and at the inner surface of each plate, and $p_{\rm out}$ is the same at the outer surface of each plate and at the faces $z = \pm L/2$ (for $L \to \infty$ it is the free-vacuum value). With the cutoff e^{-k/k_m} the difference $p_{\rm in} - p_{\rm out}$ equals -0.041043, -0.041110, -0.041120 in units of $\hbar c/a^4$ for $k_m a = 20$, 50, 100 and tends to $-\pi^2/240 = -0.041123$. The sign means attraction; with the Casimir energy density $u = -\pi^2\hbar c/720a^4$ this is $p = 3u$ [6].

12. Limitations and self-critical remarks.

(a) *Appendix B of [1].* There the pressure on the xy face was computed as $\langle H_x^2 + H_y^2\rangle/8\pi$. For electric (TM) modes the term $-\langle E_z^2\rangle/8\pi$ is missing. With it the ratio of pressure to energy density is k_z^2/k^2 for both mode types, not 1 for the electric mode (numerically for the mode $(2,1,3)$: 0.2119 instead of 1.0000). Therefore the value 1 for the electric mode, and the inequality $F/S \ge \delta(E/V)$ based on it, in the section "On the incompleteness of the standard momentum balance" must be revised. The Hamiltonian derivation of Appendix C is not affected: by item 7 it coincides with the tensor result mode by mode.

(b) *Regularization.* With a cutoff $f(k/k_m)$ the derivative of the mode energy is $-\partial_h\left[\frac{\hbar c k}{2}f\right] = \frac{\hbar c}{2h}\frac{k_z^2}{k}\left[f + \frac{k}{k_m}f'\right]$, whereas the regularized tensor sum $\sum(k_z^2/k)f$ lacks the second term. For two plates the difference disappears as $k_m a \to \infty$ (item 11). For the honeycomb at finite ak_m this has not been computed; this term may be related to the dependence $R(ak_m)$. Which of the two regularized quantities is physical cannot be decided within the perfect-conductor model with a mode cutoff; a real $\varepsilon(\omega)$, as in [8], is needed.

(c) *Cell height.* The calculation assumes that the cell walls reach the cube face, so that each cell is a closed resonator. For real cells of height $h \ll L$ the cell is open into the large

cavity, the modes do not separate, and this calculation does not describe that case.

(d) *Perfect conductor.* The jump is a property of the perfect conductor. In a real metal T_{zz} decays to zero over the skin depth, and the integral of the force over the layer equals the jump as long as the current in the metal is consistent with the field: $\mathbf{j} = \frac{c}{4\pi}\nabla \times \mathbf{H}$.

(e) *Wall thickness.* Cell walls of finite thickness carry no z-force ($T_{zx} = 0$ on them) but reduce the fraction of the plate area occupied by the cavities.

(f) *Quantum average.* The equality of electric and magnetic parts of the mode energy in the ground state (item 7) is used as a standard oscillator result and is not derived here.

(g) *What has been checked.* The fields and averages were checked numerically for individual modes, and the limiting value for two plates. The sum $\sum k_z^2/k$ for the honeycomb plate, regularized with the Euler–Maclaurin formula of [1], has not yet been evaluated.

THE STRESS TENSOR FROM THE ZERO-POINT MODE SUM

In [1] the force was obtained from the zero-point energy $\frac{1}{2}\sum\hbar\omega$ evaluated with the two-dimensional Euler–Maclaurin formula; the Maxwell stress tensor itself was not derived there. It follows from the same mode sum.

One mode. For a plane wave with wave vector $\mathbf{k}$ and energy density u the momentum flux is $\Pi_{ij} = u k_i k_j/k^2$ (Tamm's tension is $T_{ij} = -\Pi_{ij}$). A standing wave is the sum of the waves with $\pm k_z$ and has the same Π_{zz}. A zero-point mode in a cavity of volume V has $u = \hbar\omega/2V$, hence

$$\langle \Pi_{zz} \rangle = \frac{\hbar c}{2V} \sum_{n_x,n_y,n_z} \frac{k_z^2}{k}.$$

This differs from the energy sum, eq. (1) of [1], $E = \frac{1}{2}\sum\hbar\omega = \hbar c \frac{1}{2}\sum_{n_x}\sum_{n_y}\sum_{n_z} k$, only by the summand: k_z^2/k instead of k.

Link to the Hamiltonian. With $k_z = \pi n_z/l$ one has $\partial k/\partial l = -k_z^2/(kl)$, so for every mode separately

$$-\frac{\partial}{\partial l}\frac{\hbar c k}{2} = \frac{\hbar c}{2l}\frac{k_z^2}{k} = S\Pi_{zz}^{(\mathrm{mode})}, \qquad V = Sl.$$

Thus $-\partial E/\partial l = S\Pi_{zz}$ holds mode by mode (the Hellmann–Feynman theorem [5]; equivalently, conservation of the adiabatic invariant E/ω of each mode under a slow displacement of the wall). The Hamiltonian force $\partial\langle H\rangle/\partial l$ of the preceding sections and the stress-tensor force therefore coincide already before regularization.

Regularization. The cutoff function $f(k/k_m)$ and the Euler–Maclaurin procedure of [1] apply unchanged, with the function G built from k_z^2/k. At finite k_m the two expressions may differ by terms containing $f'(k/k_m)$, because f depends on $|\mathbf{k}|$. Whether these terms are the origin of the material dependence $R(ak_m)$ remains to be checked by calculation.

Symmetry check. The renormalized $\langle T_{\mu\nu}\rangle$ of the electromagnetic field is traceless. Between two parallel plates the boundary conditions are invariant under boosts along the plates; together with the trace condition this gives $\langle\Pi_{zz}\rangle = 3\langle u\rangle$ (Brown and Maclay [6]), which is the factor $F/S = -3\,\delta(E/V)$ found for Casimir's configuration in [1]. A honeycomb cell is a waveguide along z: away from its ends its boundary conditions are invariant under boosts along z, so in the (t,z) block $\langle T_{\mu\nu}\rangle \propto \eta_{\mu\nu}$ and $\langle\Pi_{zz}\rangle = -\langle u\rangle$. This reproduces $|F/S| = \delta(E/V)$ of Appendices B and C of [1]. The argument is exact for a perfect conductor in the limit $ak_m \to \infty$ and up to end corrections of order a/l; the material cutoff breaks the boost invariance, and its corrections require the mode sum above.

The same flux can be computed without a mode cutoff by the method of Dzyaloshinskii, Lifshitz and Pitaevskii [8], who obtain the stress in a vacuum gap from the Green function of the field, the material entering through $\varepsilon(\omega)$. For a honeycomb cell this requires the Green function of a rectangular waveguide and is left for future work.

ON THE AUXILIARY CAVITY AND THE POSITION OF THE PILLBOX

The perfectly conducting walls of the cube L^3 (Fig. 2) are, as in Casimir's paper, an auxiliary boundary condition without a physical counterpart, in the same way as the periodic boundary conditions used in textbooks of quantum mechanics to normalize plane waves. The procedure is legitimate because the result is taken in the limit $L \to \infty$. By Weyl's theorem [7] the leading term of the number of modes, $N(k) \simeq Vk^3/3\pi^2$, does not depend on the boundary conditions on the outer faces, which affect only surface corrections, and the contribution of the remote walls to the force on the plate vanishes as $L \to \infty$. The finite box makes the wave numbers discrete; it does not make the space discrete. The cutoff k_m is a separate, physical ingredient: it is set by the transparency of the plate material for short waves. (If the vacuum were regarded as a discrete medium with lattice step b, its Brillouin zone would add a cutoff $|\mathbf{k}| \le \pi/b$; for the force on the plate, however, the relevant cutoff is that of the plate material.)

Position of the pillbox faces. Take the faces at $z = \pm d$, $0 < d < L/2$, strictly inside the two cavities. There is no matter inside a cavity, so $\partial_k\langle T_{zk}\rangle = 0$. Integrated over the cross-section of a cell, the flux through the side walls vanishes: on a perfect conductor $E_z = 0$ (tangential) and $H_n = 0$ (normal), so $T_{zn} = (E_zE_n + H_zH_n)/4\pi = 0$. Hence $\int\langle T_{zz}\rangle\,dx\,dy$ does not depend on z, and F_z does not depend on d. The faces at $z = \pm L/2$ used above are to be understood as the limit $d \to L/2$ taken from inside the cavities; at $z = \pm L/2$ themselves stand the auxiliary conducting walls. On the honeycomb side the face crosses the cell walls; for perfectly conducting walls of

zero thickness these do not contribute to the integral.

ON THE INCOMPLETENESS OF THE STANDARD MOMENTUM BALANCE

It is essential to specify which momentum is not conserved, so that the result is not misread as a violation of a fundamental conservation law. The standard momentum-balance equation of continuum electrodynamics (Tamm, § 105) reads

$$\nabla \cdot \mathbf{T} = \frac{\partial \mathbf{g}}{\partial t} + \mathbf{f}_{\rm vol}, \qquad \mathbf{g} = \frac{1}{c^2}\mathbf{S} = \frac{1}{4\pi c}[\mathbf{E}\times\mathbf{H}],$$

where $\mathbf{T}$ is the Maxwell stress tensor, $\mathbf{g}$ the field momentum density and $\mathbf{f}_{\rm vol}$ the volume (ponderomotive) force density. For the vacuum fluctuations of a static honeycomb system the mean Poynting vector vanishes, $\langle \mathbf{S} \rangle = 0$, hence $\langle \partial \mathbf{g}/\partial t \rangle = 0$.

The customary identification of the field momentum with $\mathbf{g} = \mathbf{S}/c^2$ therefore misses the contribution that remains: the surface integral of the stress tensor over the boundary of the vacuum region.

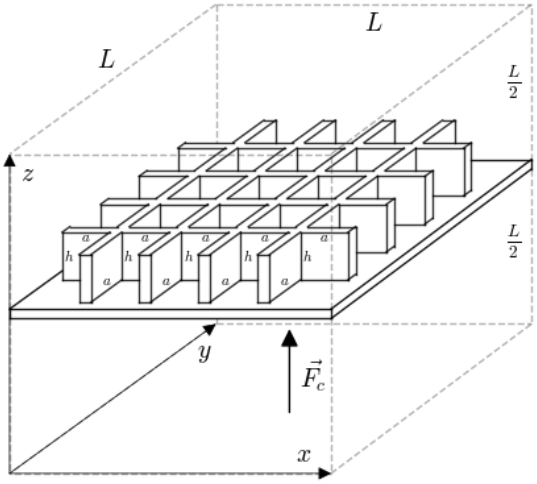


Fig.2. Cubic cavity with a plate covered by honeycomb.

Consider a closed pillbox surface $\partial V_{\rm vac}$ whose faces lie in the free vacuum at $z = \pm L/2$ on either side of the plate (the plate is entirely enclosed, and the lateral faces cancel by in-plane symmetry).

The net force is

$$F_z = \oint_{\partial V_{\rm vac}} \langle T_{zz} \rangle \, da = \left[\langle T_{zz} \rangle_{z=+L/2} - \langle T_{zz} \rangle_{z=-L/2} \right] S.$$

Because the faces lie in the vacuum, this is the flux of the field momentum through a purely vacuum boundary, not a force applied directly to the substance of the plate. The difference is non-zero solely because the mode structure — and hence $\langle T_{zz} \rangle$ — differs on the honeycomb and the smooth side. This is precisely the term omitted when the balance is reduced to $\mathbf{g} = \mathbf{S}/c^2$ with $\langle \mathbf{S} \rangle = 0$.

Quantitatively, the surface integral reproduces the thrust obtained earlier from the difference of specific energy densities [1]: for a gold nano-honeycomb with cell size $a \approx 2\ \mu$m,

$$\frac{F}{S} = \langle T_{zz} \rangle_{\rm honey} - \langle T_{zz} \rangle_{\rm smooth} \approx 8.55\ \text{dyn/m}^2 = 8.55 \times 10^{-5}\ \text{N/m}^2,$$

of the order $\langle T_{zz} \rangle \sim K\hbar c/a^4$ with a dimensionless geometric coefficient $K \sim 4 \times 10^{-2}$ set by the cell mode structure. Thus the § 105 reading does not re-derive the force but interprets it: the same number is the flux of field momentum through the vacuum boundary.

Where the standard derivation is incomplete.

The passage from volume forces to stresses rests on the Gauss theorem, which Tamm applies (§ 33, eq. 33.7) under the explicit condition that the components of the stress tensor are continuous inside the volume; surfaces of discontinuity are excluded there, and only their surface charges are treated later (§ 34, eq. 34.8). A vacuum–metal boundary of a resonator can carry zero surface charge and still produce a finite jump of the stress tensor: the tensor evaluated with the fields on the vacuum side differs from the one evaluated with the fields inside the wall. On the two sides of such a surface element the stresses are then not equal and opposite. This case is not considered in § 33, nor anywhere in Tamm's book, including § 105. For a volume V that contains internal discontinuity surfaces S_j the Gauss theorem has to be written as

$$\int_V \frac{\partial T_{ik}}{\partial x_k}\,dV = \oint_{\partial V_{\rm ext}} T_{ik}\,n_k\,dS \;-\; \sum_j \int_{S_j} [\![\,T_{ik}\,]\!]\,n_k\,dS_j,$$

where $[\![\cdot]\!]$ is the jump across S_j. The customary identification of the field momentum with $\mathbf{g} = \mathbf{S}/c^2$ keeps only the external surface integral and drops the jump terms.

The jump on a honeycomb plate. The plate has two such internal boundaries: the bottoms of the cells on the structured side and the smooth face on the other side. The stress on each of them is the electromagnetic pressure of the vacuum modes. Appendix B of [1] computes this pressure on the xy face of an $a\times b\times h$ rectangular resonator, $f_z/S = \int (H_x^2 + H_y^2)\,dS/(8\pi S)$, mode by mode, and relates it to the mode energy density E/V. For a wave propagating along z

$$\frac{f_z/S}{E/V} = 1 \quad \text{(electric mode, } H_z = 0\text{)}, \qquad \frac{f_z/S}{E/V} = \frac{a^2 k_z^2}{a^2 k_z^2 + \pi^2 (n_x^2 + n_y^2)} \quad \text{(magnetic mode, } E_z = 0\text{)},$$

and for every mode type $\lim_{k_z\to\infty}(f_z/S)/(E/V) = 1$. Summed over the zero-point modes, the jump of $\langle T_{zz}\rangle$ across each boundary is therefore fixed by the vacuum energy density of the adjacent cavity, and the difference of the two jumps gives

$$\frac{F}{S} = [\![\langle T_{zz}\rangle]\!]_{\rm honey} - [\![\langle T_{zz}\rangle]\!]_{\rm smooth} \;\geq\; \delta\frac{E}{V} = \hbar c\,\pi\frac{R}{a^4},$$

the inequality of [1] (Appendix B). It coincides with the Hamiltonian result $\partial\langle H\rangle/\partial l$ of the previous section. For a gold nano-honeycomb with cell size $a \approx 2$ μm this gives $F/S \approx 8.55$ dyn/m$^2 = 8.55\times 10^{-5}$ N/m^2. The § 105 reading, completed by the jump terms, therefore does not re-derive the force but locates it: the unbalanced momentum sits in the jump of the vacuum stress tensor on the internal boundaries, which the reduction to $\mathbf{g} = \mathbf{S}/c^2$ drops.

Equivalent pillbox form. Enclosing the whole plate by a pillbox $\partial V_{\rm vac}$ with faces in the free vacuum at $z = \pm d$ (lateral faces cancel by in-plane symmetry) turns the sum of the jumps into an external flux, $F_z = \oint_{\partial V_{\rm vac}} \langle T_{zz}\rangle\,da = \big[\langle T_{zz}\rangle_{z=+d} - \langle T_{zz}\rangle_{z=-d}\big]S$. This is the flux of field momentum through a purely vacuum boundary, non-zero solely because the mode structure, and hence $\langle T_{zz}\rangle$, differs on the honeycomb and on the smooth side.

Two remarks make the interpretation unambiguous. First, it is the momentum of the field (equivalently, of the material plate that absorbs it), not the total momentum of the closed laboratory system, that is not conserved; the total momentum of plate, field and support is conserved, exactly as for an ordinary sail in a wind. Second, the renormalised vacuum stress $\langle T_{\mu\nu}\rangle$ is a physically measured quantity — the Casimir effect itself — so the jump terms require no assumption of a material aether: their value is fixed by the same $\langle T_{\mu\nu}\rangle$ that is confirmed experimentally for the symmetric configuration. The broken translational symmetry of the boundary conditions, not of space itself, is what renders $\partial\langle H\rangle/\partial l \neq 0$.

CONCLUSION

The Casimir honeycomb drive exemplifies a physical system where Noether's theorem directly explains the origin of macroscopic force: **the lack of spatial homogeneity leads to non-conservation of momentum**, and hence to the appearance of thrust. This demonstrates that quantum vacuum effects are not mere mathematical artifacts but manifestations of deep symmetry principles. Experimental verification can not only confirm the reality of virtual photons but also validate the link between geometry, symmetry, and dynamics in quantum field theory.

ТЕОРЕМА НЕТЕР ТА СТИЛЬНИКОВИЙ РУШІЙ КАЗИМИРА: ПРО НЕЗБЕРЕЖЕННЯ ІМПУЛЬСУ В АСИМЕТРИЧНИХ ВАКУУМНИХ СИСТЕМАХ

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У цій роботі аналізується стильниковий рушій Казимира через призму теореми Нетер. Розглянута система — ідеально провідна пластина з нанорозмірною сотовою структурою з одного боку — не має трансляційної симетрії вздовж осі, нормальної до пластини, через геометричну асиметрію вакуумних мод з двох її сторін. Як наслідок, гамільтоніан вакууму явно залежить від координати l (ефективної довжини резонатора), і, згідно з теоремою Нетер, відповідна компонента імпульсу не зберігається. Це не збереження проявляється як результуюча сила квантового вакууму (тяга Казимира). Показано, що отриманий вираз для сили на одиницю площі безпосередньо відображає цю порушену симетрію й узгоджується як з гамільтоновим підходом, так і з різницею густин енергії. Результат підкреслює глибокий фізичний принцип: макроскопічна тяга від квантових флуктуацій вакууму виникає саме тому, що система не є інваріантною відносно просторових зсувів. Окремо показано, що стандартне застосування рівняння балансу імпульсу (§ 105 Тамма) є неповним для систем із $\langle \mathbf{S} \rangle = 0$: оскільки середній вектор Пойнтінга дорівнює нулю, незбалансований імпульс міститься у поверхневому інтегралі тензора натягів Максвелла по межі вакуумної області, а не у потоці $\mathbf{g} = \mathbf{S}/c^2$. Не зберігається імпульс поля (речовини), а не повний імпульс замкненої системи; величина сили спирається на виміряний вакуумний тензор $\langle T_{\mu\nu} \rangle$ і не потребує припущення про матеріальний ефір.

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